SIEMENS Healthineers

MAGNETOM

Troubleshooting Guide System

RF-System

Aera
Skyra
Avanto fit
Skyra fit
Prisma
Prisma fit

Document Version

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1.1 Notes

1.1.1 Safety Information and/or references to Safety Information

Adhere to ESD regulations and use the appropriate tool set: Anti-static wristband and grounded pad. (TDK equipment type 8005 from 3M Deutschland GmbH).

WARNING

Please follow the general safety guidelines (→ General Safety Notes / TD00-000.860.01) as well as the MR-Specific Safety Information (→ MR-specific safety information / MR-000.860.01)!

Failure to comply with the Safety Information may result in death or serious injury.

» **Read the Safety Information contained in the general safety notes as well as the MR-Specific Safety Information in detail prior to starting work. Adhere to the information provided at all times.**

CAUTION

Strong magnetic field!

Magnetic objects may cause injury.

» **Maintain a safe distance to the magnetic field.**

CAUTION

High voltage!

» **Before opening RFCEL, switch off power for RFCEL (circuit breaker F10/EPC) and for PTAB (circuit breaker F24/EPC).**

» **Before replacing TX-Box and/or THYRE, switch off power for TX-system (circuit breaker F51/F52 in EPC). Check that high voltage LEDs on TX-Box and THYRE are off.**

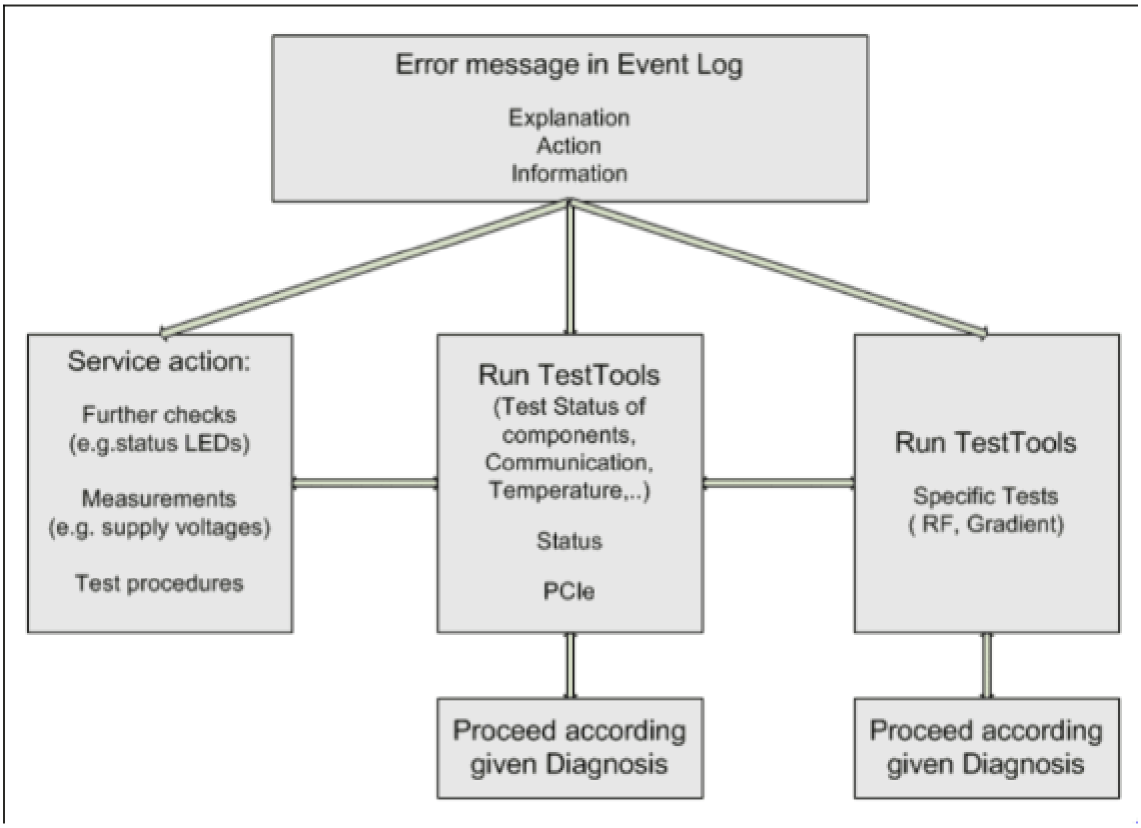
1.1.2 General Remarks

The main tool for troubleshooting consists of error messages delivered from the RF-System which are shown in the error log file (EventLog) of the Service Software. Click on the message IDs there to get further advice.

RF-components (including cables and FOCs)	Error message source (enter this string in the EventLog field "source" to filter for component-specific messages)
RFCEL_COM	MRI_CMM
RFCEL_IF_RCCS	MRI_RCS
RFCEL_RFIS (BC_DYN, LC_DYN	MRI_RFI
RFCEL_COM (DSP STIMO, PALI)	MRI_RSP
RF Safety Watchdog (RFSWD)	MRI_RFS
RFCEL_Dig_RX_32	MRI_RXV
TX-Box Controller	MRI_TBC
TX-Box RFPA	MRI_TXB
LC, BC_DYN, SW	MRI_CSS
LC, BC_DYN, RFCEL_COM	MRI_COI

1.1.2.1 General strategy for troubleshooting

Fig. 1: General proceeding in trouble-shooting



1.1.3 Product-specific Remarks

Tab. 1 1.5 T Systems

System	Aera 1.5 T	Avanto^{fit} 1.5 T
TX-Box - Type Material number	TX-Box 098 1T5 LCTX 10432773 **(only for system serial numbers 41005 - 41212 and 52001 - 52025)** HF transmitter TXB_2 1,5T 10683697 **(only for system serial numbers 41213 - 44499 and 52026 - 53499)**	TXB_2 1,5T 10683697
Body Coil - Type Material number	Body Coil 1.5 T 10358977	Body Coil Avanto^{fit} 1.5 T 10676455

Tab. 2 3T Systems

System	Skyra 3T	Skyra^{fit} 3T	Prisma, Prisma^{fit}
TX-Box - Type Material number	TX-Box 098 3T TF LCTX 10432717 **(only for system serial numbers 45005 - 45178 except systems with MNO-Option)** HF transmitter TXB_2 3T 10683698 **(only for system serial numbers 45179 - 48499 and systems with MNO-Option)** TBX_2CH 3T 10432771 **(only for systems with "TIM TX True Shape" option)**	TXB_2 3T 10683698 TBX_2CH 3T 10432771 **(only for systems with "TIM TX True Shape" option)**	TBX_2CH 3T 10432771
Body Coil - Type Material number	Body Coil 3 T 1035978 Body Coil Skyra 2ch 10676432	Body Coil Skyra^{fit} 3T 10676447	Bodycoil 021 10676427

1.2 Requirements

1.2.1 Additionally Required Documents

Function Description "RF-System"

Replacement of Parts "RF-System"

Error messages RF-System → Healthcare Services Knowledge Base (SKB)

Wiring Diagrams

1.2.2 Required Aids and Tools

- Spare FOCs
- Non-magnetic tool set
- DVM
- Service Plug Aera/Skyra complete (material number 10500104 with Rev. 2 or higher)

2.1 General

This section describes procedures for troubleshooting components (FRUs) of the RF-System.

2.2 TX-Box

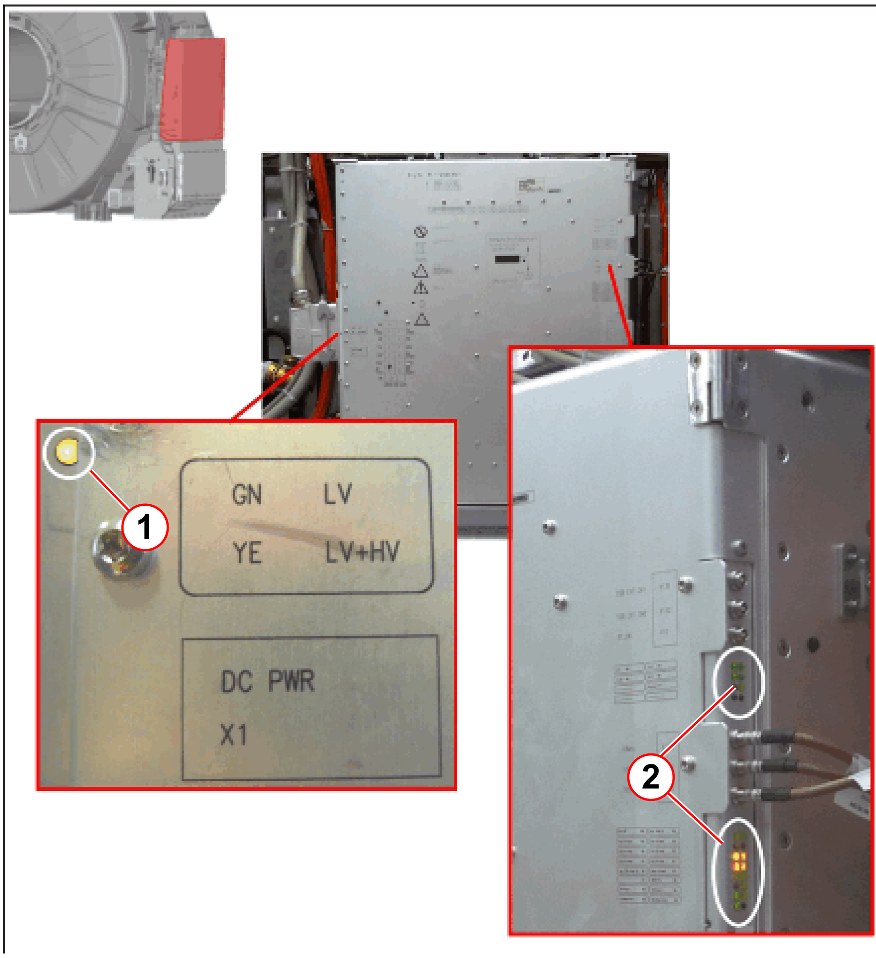
The description is valid for the following TX-Box versions:

- TBX1_ 1.5T (10432773)
- TBX1_ 3T (10432717)
- TBX2_ 1.5T (10683697)
- TBX2_3T (10683698)
- TBX2ch_pTX_3T (10432771)

2.2.1 LEDs

2.2.1.1 Overview

Fig. 2: Location of TX-Box LEDs

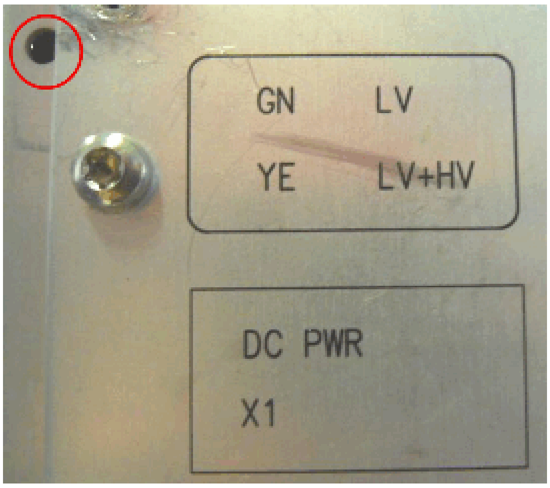


(1) LED left-side (at PoliFi power connector)

(2) LEDs right-side

2.2.1.2 All TX-Box Versions - High and Low DC-voltage LEDs at Power Line Filter (POLIFI)

Fig. 3: High and low DC-Voltage LED, TXB left-side



During normal boot, this LED turns-on from dark directly to yellow, as DC low voltage and high voltage come up at the same time.

The printed LED label does not exist at TBX1.

Tab. 3 DC-power LEDs, all TXBs

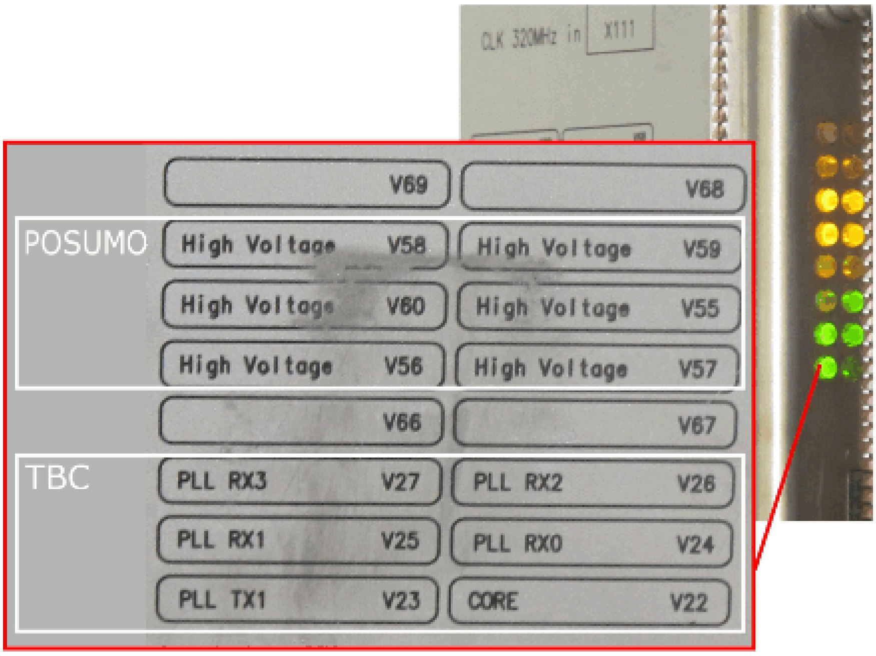
LED	Location	State	Label
V	DC-power connector	green - DC low voltage present	GN LV

		Yellow - DC low voltage and high voltage present
		
		`Note:`
		`at most of the old TXB_1 versions LED is red!`
			YE LV+HV

		off - low/no voltage	n.a.

2.2.1.3 TX-Box1 - High DC-voltage LEDs at Power Supply Module (POSUMO) and communication LEDs (TBC)

Fig. 4: High DC-voltage and communication LEDs, TXB1 right-side

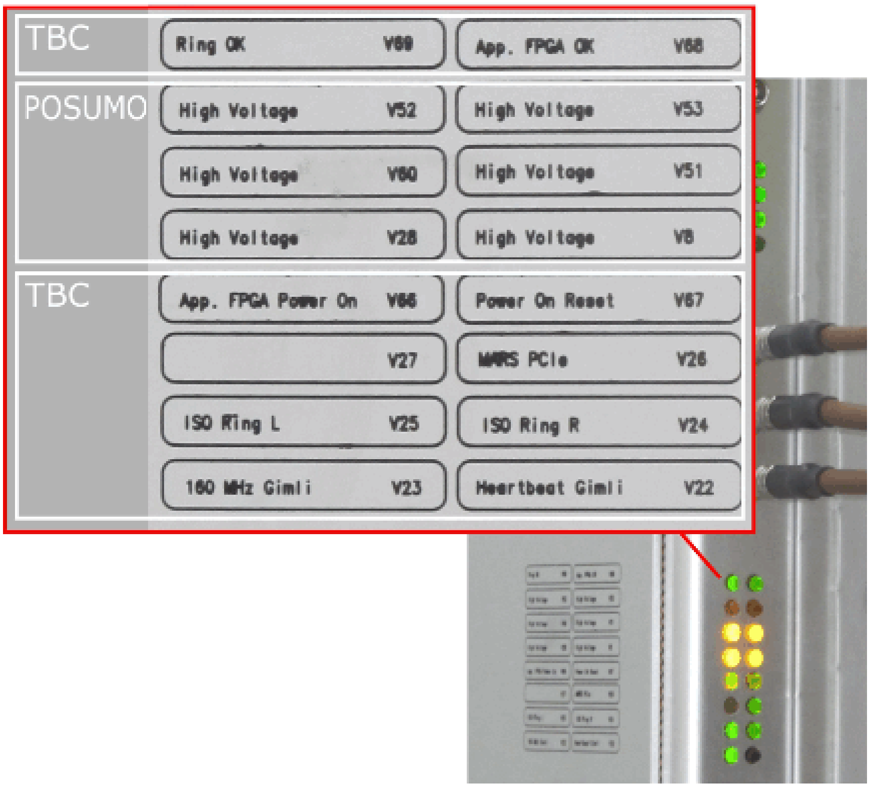


Tab. 4 High DC-voltage and communications LEDs, TXB1 right-side

LED	Location	State	Label
V22	TXB right-side	blink - ok red - error	CORE (Heartbeat GIM-LI)
V23		green - ok red - error	PLL TX1 (160MHz GIMLI)
V24		green - ok red - error	PLL RX0 (ISO Ring R)
V25		green - ok red - error	PLL RX1 (ISO Ring L)
V26		green - ok red - error	PLL RX2 (MARS PCIe)
V27		off	PLL RX3 (none)
V55		on - voltage present (3T only)	High Voltage
V56		on - voltage present (1.5T+3T)	High Voltage
V57		on - voltage present (1.5T+3T)	High Voltage
V58		- -	High Voltage (not used)
V59		- -	High Voltage (not used)
V60		on - voltage present (3T only)	High Voltage
V66- V69		- -	none

2.2.1.4 TX-Box2 - High DC-voltage LEDs at Power Supply Module (POSUMO) and communication LEDs (TBC)

Fig. 5: High DC-voltage and communication LEDs, TXB2 right-side

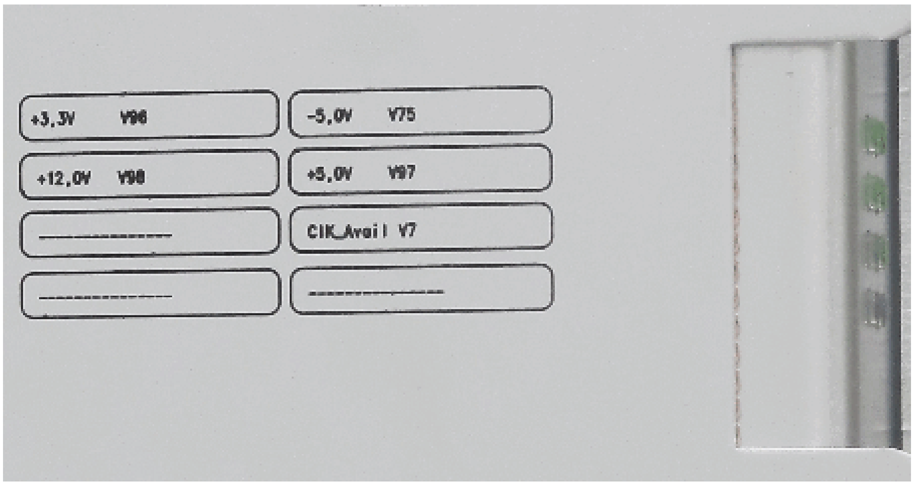


Tab. 5 High DC-voltage and communications LEDs, TXB2 right-side

LED	Location	State	Label
V8	TXB right-side	on - voltage present (1.5T+3T)	High Voltage
V22		blink - ok red - error	Heartbeat Gimil
V23		green - ok red - error	160 MHz Gimli
V24		green - ok red - error	ISO Ring R
V25		green - ok red - error	ISO Ring L
V26		green - ok red - error	MARS PCIe
V27		- -	none (not used)
V28		on - voltage present (1.5T+3T)	High Voltage
V51		on - voltage present (3T only)	High Voltage
V52		- -	(not used)
V53		- -	(not used)
V60		on - voltage present (3T only)	High Voltage
V66		green - ok red - error	App. FPGA Pwr On
V67		green - ok red - error	Power On Reset (Gimli on)
V68		green - ok red - error	Appl. FPGA OK
V69		green - ok red - error **Note:** LED V69 (Ring OK) will only reset after Meas&Recon reboot.	Ring OK (Emergency Fault)

2.2.1.5 TX-Box2 - Internal Low Voltages and Clock

Fig. 6: Common internal low voltage and clock LEDs, TXB right-side



Tab. 6 DC-power LEDs; all TXBs

LED	Location	State	Label
V7	TXB right-side	green - ok (Clock available) red - error	CIK_Avail
V75		green - ok red - error	-5,0V
V96		green - ok red - error	+3,3V
V97		green - ok red - error	+5,0V
V98		green - ok red - error	+12,0V

2.2.2 Check of Supply Voltages TX-Box

WARNING

Dangerously high voltage can be present even when the MR system is off.

Risk of electrical shock!

» **Before start working at the TX-Box, switch off the power for the TX-system (circuit breaker F51/ F52 in EPC).**

» **Wait at least 10 min. and check for no voltage at the THYRE output.**

» **Do not remove any cables before the high voltage LEDs at THYRE and the TX-Box are off.**

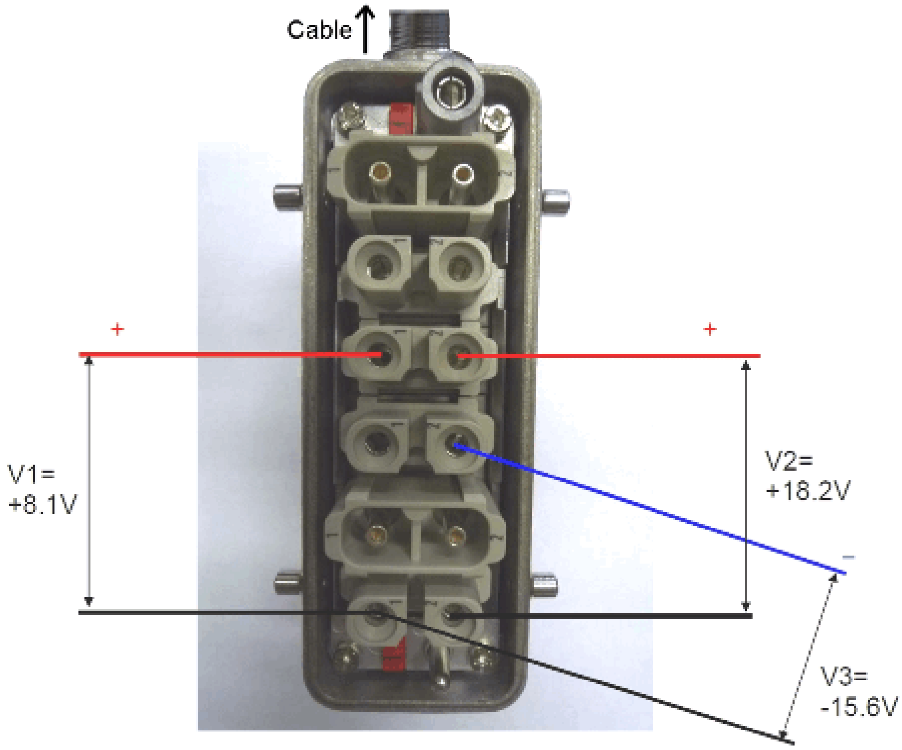
2.2.2.1 110/130V DC Voltage at TX-Box

The DC High Voltage for the TX-Box can be measured at the THYRE output, see (→ Check of Supply and Output Voltages THYRE / Page 39).

2.2.2.2 Measurement of Open Circuit DC Low Voltages at TX-Box

- Select **System/ Control/ Meas Recon/ Standby**
- At the EPC, switch off circuit breakers F51/F52/ THYRE and F10/ RFCEL/TX-Box.
- Wait at least 10 min. and **check for NO VOLTAGE** at the THYRE output.
- Check that the high voltage LEDs at THYRE and the TX-Box are off.
- Unplug the Power Line Filter (PoliFi) plug from the left side of the TX-Box
- At the EPC, switch ON circuit breaker F10/ RFCEL/TX-Box (TX-Box low DC-voltages).
- Switch the MR system “ON” at the Alarm Box
- Measure the low DC-voltages at the PoliFi plug pins, see next figure (→ Fig. 7 Page 18).
- When measurements are finished, switch off circuit breaker F10/ RFCEL/TX-Box again and restore original configuration.

Fig. 7: PoliFi plug removed from TXB



2.2.2.3 Measurement of Open Circuit DC Voltages at Low Voltage Power Supply

Fig. 8: LVPS at Filter Plate (connectors removed)

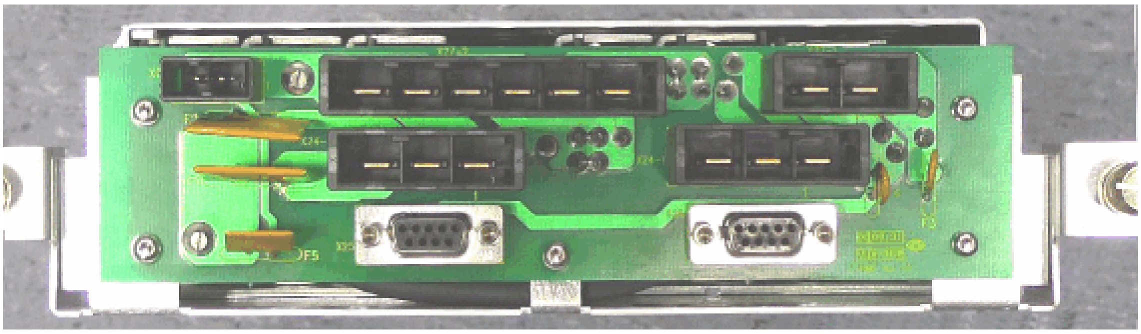
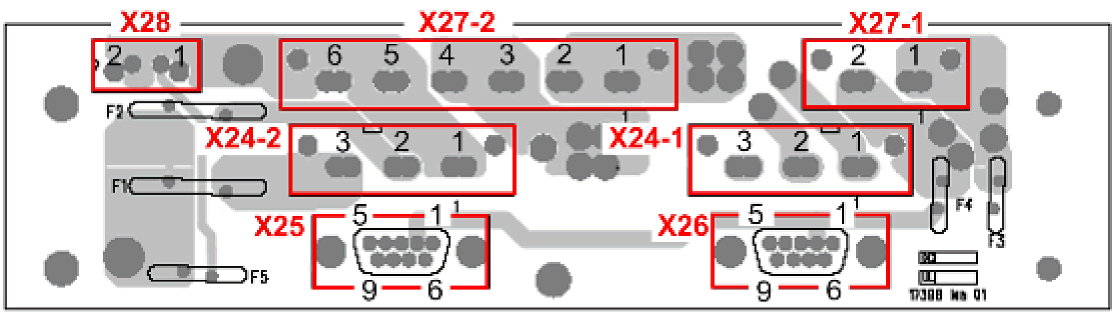


Fig. 9: LVPS connector layout



- Select **System/ Control/ Meas Recon/ Standby**
- At the EPC, switch off circuit breakers F10/ RFCEL/TX-Box and F24/ PTAB.

- At equipment room side, remove the connector(s) at the Low Voltage Power Supply (LVPS).
- At the EPC, switch ON circuit breakers F10/ RFCEL/TX-Box and F24/ PTAB.
- Switch the MR system "ON" at the Alarm Box.
- Measure the voltages with DVM at the connector contact pins (at piggy-back board front-side, see (→ Fig. 8 Page 18).
- Switch off circuit breakers F10/ RFCEL/TX-Box and F24/ PTAB. Restore original configuration.

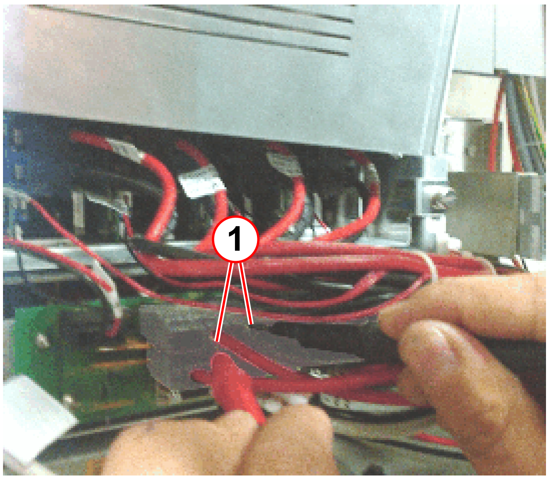
The supply voltages for the TX-Box are located at connectors X27-1 and X27-2.

Tab. 7 LVPS test points

Connector	Pin	Description
X24-1	1	GND (-32 V)
	2	-32 V @ 8 A
	3	n/c
X24-2	1	GND (26.6 V)
	2	+26.6 V @ 8 A
	3	+26.6 V @ 6 A
X25	2	GND (18.2 V)
	3	+18.2 V
X26	2	GND (18.2 V)
	3	+18.2 V
X27-1	1	+18.2 V @ 13 A
	2	GND (18.2 V)
X27-2	1	+8.1 V @ 22 A
	2	+8.1 V @ 22 A
	3	GND (8.1 V/-15.6 V)
	4	GND (8.1 V/-15.6 V)
	5	-15.6 V @ 6 A
	6	n/c
X28	1	+26.6 V @ 4A
	2	GND (26.6 V)

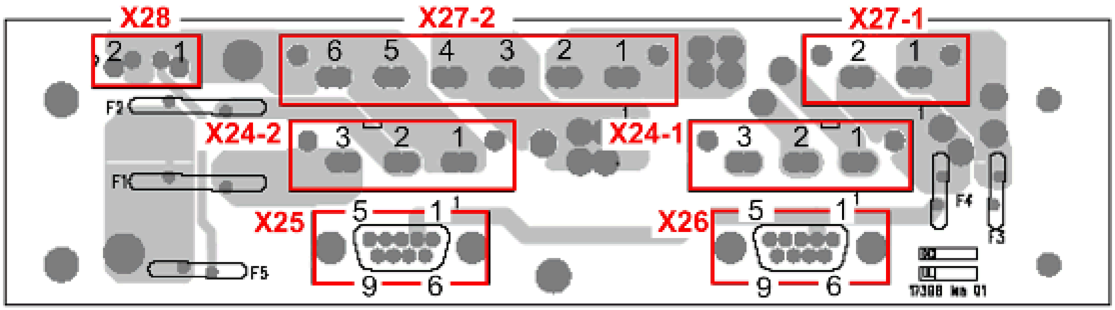
2.2.2.4 Measurement of load condition DC Voltages at Low Voltage Power Supply

Fig. 10: LVPS at filter Plate (connectors fitted)



(1) Measure with DVM at fitted connector

Fig. 11: LVPS connector layout



- At the RF-filter plate, equipment room side, measure the voltages with a DVM at the LVPS.
- Measure the voltages at the pins of the fitted connectors, see (→ Fig. 10 Page 20).

> The supply voltages for the TX-Box are located at connectors X27-1 and X27-2.

Tab. 8 LVPS test points

Connector	Pin	Description
X24-1	1	GND (-32 V)
	2	-32 V @ 8 A
	3	n/c
X24-2	1	GND (26.6 V)
	2	+26.6 V @ 8 A
	3	+26.6 V @ 6 A

Connector	Pin	Description
X25	2	GND (18.2 V)
	3	+18.2 V
X26	2	GND (18.2 V)
	3	+18.2 V
X27-1	1	+18.2 V @ 13 A
	2	GND (18.2 V)
X27-2	1	+8.1 V @ 22 A
	2	+8.1 V @ 22 A
	3	GND (8.1 V/-15.6 V)
	4	GND (8.1 V/-15.6 V)
	5	-15.6 V @ 6 A
	6	n/c
X28	1	+26.6 V @ 4A
	2	GND (26.6 V)

2.2.2.5 Measurement of load condition DC Voltages at RF panel feed-through filters

In case measured load condition DC voltages at the LVPS are OK (see previous (→ Measurement of load condition DC Voltages at Low Voltage Power Supply / Page 20)) but appear too low at the TX-Box (“POSUMO” errors), inspect the Faston connectors at the RF panel feed-through filters:

- Measure the load condition DC voltages at the filters/ connectors inside the Examination Room (→ Fig. 13 Page 23).
    » If voltages are significantly lower than at the LVPS, proceed with the next step.
- Pull off and reseat the corresponding Faston connectors on both sides of the RF-filter plate (→ Fig. 12 Page 22).
    » The Faston connectors must be tight (low contact resistance).

Fig. 12: Location of Filters - Faston Connectors

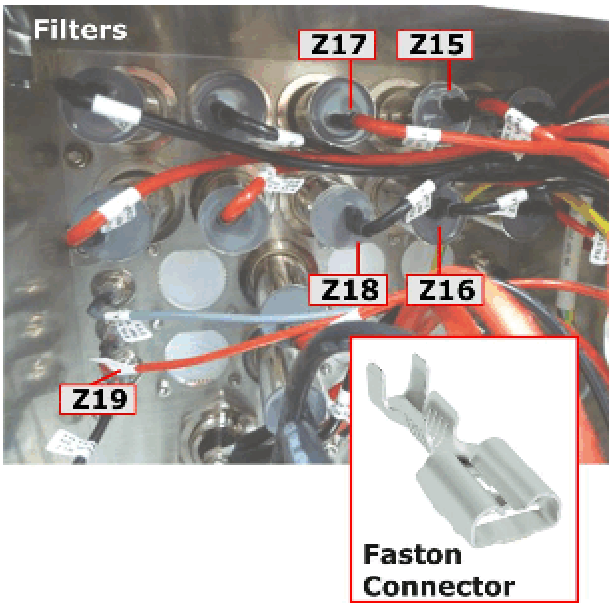
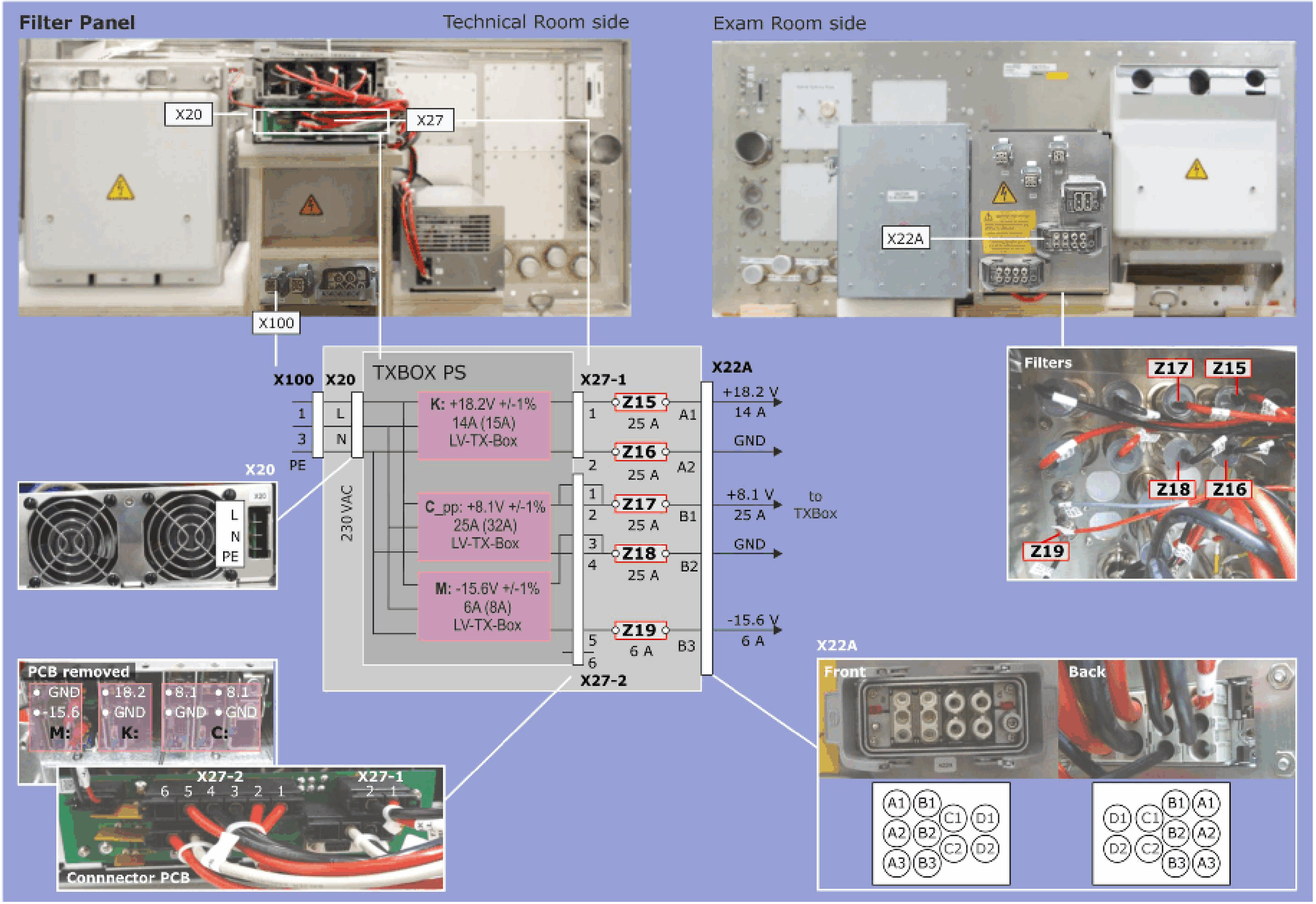


Fig. 13: Measurement points - DC Voltages at Filter Panel



2.2.3 Resolving Feedback Loop Errors

2.2.3.1 General

Products:

- Aera
- Skyra
- Prisma
- Spectra
- all fit-Upgrades

Software:

- VD13A
- VD13B
- VD13C
- VD13D
- VD13E
- VE11A
- VE11B
- VE11C
- VE11M

The TX-Box Feedback Loop (short: FBL) assures highest signal fidelity for the TX pulses by compensating deviations between the nominal and actual RF signal.

If the deviation is too large and the FBL cannot regulate the signal within the given limits, an error **MRI_TBC [164] TXOFFSOURCE_REGLERFEHLER** accompanied by **MRI_TBC [100]** or (on 2ch systems) **MRI_TBC [81]** occurs.

This can happen by sudden load changes, due to:

- sparking, e.g. defective capacitors or PIN diodes in the Body Coil or a local coil
- loose or broken cables, e.g. local TX-cable or PIN diode cabling
- TBX-internal defect, e.g. DiCo measurement receiver or CoSpli sparking
- (de-)tuning while RF is transmitted, e.g. an unconnected local coil inside the bore where the passive detuning circuit gets active.
- ...

> i
>
> Unconnected local coils will lead to TX-Box Feedback Loop Regulator errors. This problem cannot be found in the Savelog.
>
> Please inform the customer to carefully connect all coils which are placed on the patient table.

Be sure that the root cause is really a FBL error. Check the EventLog for other errors:

- A FBL error can also occur, if the TXB Power Supply Module (POSUMO) switches off due to a sequence overload. Be sure that the FBL error is not accompanied by a **MRI_TXB [31] (POSUMO_ERROR)** .
- If the TBX power supply fails there are a lot of error messages – one can be the regulator error **MRI_TBC [164]**.

2.2.3.2 Permanent / Reproducible FBL Error

If it is a permanent / reproducible error, the following procedure helps to identify the component causing the FBL error:

- Remove all coils except the transmit coil (Body Coil or local transmit coil).
  » If issue is not solved, continue with next step (bullet point).
  » If issue is solved, either one of the removed coils (incl. coil interfaces) or the corresponding coil socket, cabling or a LC_DYN is defective. Check the following:
  - Test coils on different sockets (if possible).
  - Run Coil QA.
  - Run TestTools / Interactive / LC Plug
  - Swap / replace LC_DYN boards.
- Run QA / Feedback Loop Calibration Check.
  » If QA is out of spec, re-run TuneUp / Feedback Loop Calibration. If repeatedly, QA / Feedback Loop Calibration Check runs out-of-spec, a defect in the BC might be the root cause.
- Test both, BC and local transmit.
  » If both the BC and the local transmit path show FBL errors, most probably the TBX is defective.
- Run TestTools Expert / TX / TX Phasecheck
  » If phase check is out-of-spec, the TBX might be defective.
- Run QA / BC Tuning Check and BC Image Check
  » If out-of-spec and it is hard to adjust the BC, the BC might be defective causing FBL errors.
- Move table to 3 different positions (head, abdomen, feet).
  » If FBL error occurs only at a specific table position, the RF-traps within the tabletop cable could cause the issue.
- Connect dummy loads (RF adapter set 73 85 896) to the TBX output (BC or LC).
  » If FBL error still occurs, the TBX is defective.
- Check the voltages of the TBX power supply located on the filterplate (see section, (→ Check of Supply Voltages TX-Box / Page 17))
- In case the root cause cannot be determined, please send a Savelog and the TestTools reports to HSC for further analysis.

2.2.3.3 Sporadic FBL Error

NOTICE

Escalate sporadic Feedback Loop errors immediately to the HSC!

Otherwise risk of unnecessary, expensive parts replacement and wasting of time.

» **Provide a SaveLog and contact HSC for support.**

» **The HSC can further analyze the data and provide specific repair instructions.**

If the FBL error is sporadic, collect all available information to possibly identify the defective component(s).

- Is the error reproducible with the customer protocols, different patient table positions and different loads?

  Try the coil setup which was used by the customer, when the error occurred.

  » If FBL error is reproducible, go to above section (→ Permanent / Reproducible FBL Error / Page 25).
- Which coils were connected in case of an FBL error?
  - Ask the customer: Were unconnected coils placed at the table top during measurement?

    (Unconnected coils are a root cause for FBL errors).
  - FBL error occurs with both “Rx only” and TxRx local coils?
  - FBL error occurs only with TxRx local coils?
  - FBL error occurs only with a specific set of “Rx only” local coils?
  - FBL error occurs even with BC only?
- Is there a “favorite” table position, when the FBL error occurs (e.g. only in Head position)?
- Which sequences / protocols are affected? Does the issue only occur with customer sequences?
- Helpful checks:
  - QA / Feedback Loop Calibration Check;
  - QA / BC Tuning Check;
  - QA / BC Image Check;
  - TestTools / Interactive / LC Plug;
  - TestTools / TX;
  - Coil-QA / Coil Check of involved local coils.

2.2.4 Special Problems

The following section contains additional troubleshooting information for specific error messages.

The following error messages often appear in case of a TX-problem. But usually, they are just a consequence or reaction to an initial problem. These status messages do not allow identification of the problem root cause:

- DISABLE_MODULATOR
- DISABLE_UNBLANK
- DISABLE_RFPA
- BIAS_DISABLE_UNBLANK
- BIAS_DISABLE_RF
- TBC_THYRE_WATCHDOG_ERROR
- RFPA is not ready for scanning
- POSUMO_POWER_DOWN

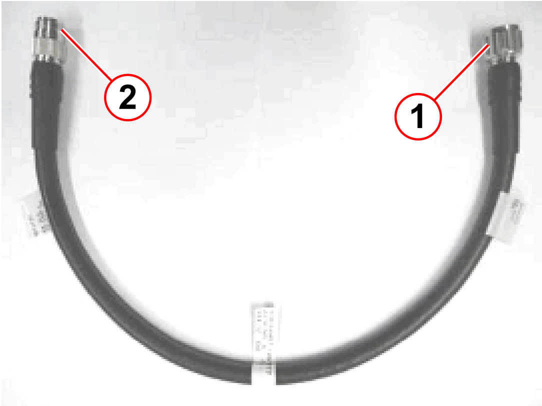
Tab. 9 Troubleshooting Procedures - Special Problems

<table>
<tr><th>Error message/
[Error code]</th><th>Possible Reason</th><th colspan="7">Action</th></tr>
<tr><td rowspan="5">MRI_RSP [E2018]
RFCELK2258_DSP Pali Error:
RF system is switched off.
Reason: UNEXPECTED RF DETECTED!

(especially while performing measurements with the Tim Coil Interface 3.0T (10500092)</td><td>Internal RF-coupling BC-TX to LC-TX path.</td><td colspan="7">Check if LC-TX cable length is correctly configured.
If incorrect, install the correct cable configuration (see information below).</td></tr>
<tr><td rowspan="2"></td><td>Avanto^{fit}</td><td colspan="2">Aera</td><td colspan="3">Skyra</td><td><b>Skyra^{fit}, Prisma, Prisma^{fit}</td></tr>
<tr><td>TBX_2</td><td>TBX_1</td><td>TBX_2</td><td colspan="2">TBX_1</td><td>TBX_2</td><td>TBX_2</td></tr>
<tr><td>Fixed Patient Table</td><td rowspan="2">no extension</td><td>Ser.no.:
41005 - 41018

UI MR026/10/P, gray cable extension fitted, see (→ Fig. 15 Page 29).</td><td>no extension</td><td>Ser.no.:
45005 - 45017
UI MR026/10/P, gray cable extension fitted, see (→ Fig. 15 Page 29).</td><td>Ser.no:
45018 - 45178

black cable extension W11777 can be fitted in case of “UNEXPECTED RF” error, see (→ Fig. 14 Page 29).</td><td rowspan="2">Ser.no:
45179 - 48499
and
45018 - 45178
(TBX_1 to TBX_2 upgrades)
no MNO option:
black cable extension W11777 delivered with TXB_2 must be fitted, see (→ Fig. 14 Page 29).
MNO option:
no extension
Tim TX True Shape opt.:
no extension</td><td rowspan="2">all Ser.no:
no MNO option:
black cable extension W11777 delivered with TXB_2 must be fitted, see (→ Fig. 14 Page 29).
MNO option:
no extension
Tim TX True Shape opt.:
no extension</td></tr>
<tr><td>Mobile Patient Table</td><td colspan="4">no extension</td></tr>
</table>

<table>
<tr><th>Error message/
[Error code]</th><th>Possible Reason</th><th colspan="2">Action</th></tr>
<tr><td rowspan="3">see before</td><td rowspan="3">Internal RF-coupling from BC-TX into LC-TX path/ wrong LC-TX cable length.</td><td colspan="2">Check if LC-TX cable length is correctly configured.
If incorrect, install the correct cable configuration (see information below).</td></tr>
<tr><td>Fig. 14: LC-TX cable extension kit W11777 (material no. 10684522)


(1) Connect to LC_TX output TXB_2
(2) Connect to LC-TX cable

<code>Note: The black extension cable is delivered with the spare part TXB_2 or can be ordered from SPC.</code></td><td>Fig. 15: LC-TX cable extension UI MR026_10/P


(1) Connect to LC-TX output TBX_1
(2) Connect HN adapter to LC-TX cable

<code>Note: The gray extension cable was fitted on some systems with UI MR026/10/P</code></td></tr>
<tr><td colspan="2"><code>Note: The combination of 2 extension cables is not allowed!</code></td></tr>
</table>

Error message/ [Error code]	Possible Reason	Action
MRI_TBC [100] TxK2304 unit error: **CH0_TX_QUEUE_REGLER_FAULT** OR on 2ch systems: **MRI_TBC [81]** TxK2304 unit error: **CH1_TX_QUEUE_REGLER_FAULT**	Mostly, defective Local Coils or Body Coil. Seldom, TX-Box is the problem.	See section (→ Resolving Feedback Loop Errors / Page 24).
MRI_TBC [164] TxK2304 unit error: **TXOFFSOURCE_REGLERFEHLER**		
MRI_TBC [216] TxK2304 unit error: CO-SPLI_Sense_**STUCK-LOW_Pulse**	For TXB with Rev. <5: TX-Box is the problem.	Check / replace TX-Box.
	For TXB with Rev. >4: RF-overload from sequence. Otherwise, TX-Box is the problem.	1. Check for wrong sequence parameters (RF-amplitudes, -timing). Run measurement with SIEMENS standard protocol. 2. Check / replace TX-Box.

Error message/ [Error code]	Possible Reason	Action
MRI_TBC [217] TxK2304 unit error: CO-SPLI_Sense_**STUCK-HIGH_Pulse**	Always, TX-Box is the problem.	Check / replace TX-Box.
MRI_TBC [220] TxK2304 unit error: CO-SPLI_Sense_**STUCK-LOW_NoPulse**	Mostly, a cooling water problem: ■ short pump failure ■ secondary pressure / flow too low ■ cooling water over-temperature. Only in rare cases, TX-Box is the problem.	1. Check water flow and temperature into TX-Box (approx. 18 l/min, 18 - 22 degC). 2. Check the EventLog of the system for warnings and errors of the cooling system, since these may be the root cause for the Tx-Box error message. 3. Check RFCEL air cooling for blocking, leakage or overtemperature. 4. Check/adjust TX-Box water pressure at gauge to approx. 3.2bar. 5. Perform visual inspection of TX coaxial cables and connectors for defects, loose or burnt connectors. 6. Run TestTools / Fast /TX, Normal / TX and Stability / TX to check BC and TX-Box. 7. Check TX-Box LEDs, see (➔ LEDs / Page 10). Check supply voltages, see (➔ Check of Supply Voltages TX-Box / Page 17).
MRI_TBC [221] TxK2304 unit error: CO-SPLI_Sense_**STUCK-HIGH_NoPulse**	Only information.	Ignore.

Error message/ [Error code]	Possible Reason	Action
MRI_TXB [2] TxK2304 unit error: **Junction overtem-perature error**. OR IN CONJUNC-TION WITH: **MRI_TXB [4]** TxK2304 unit error: Tx-Box **gate volt-age max error** without rf opera-tion.	Mostly, a cooling water problem: ■ cooling water over-tem-perature. Otherwise, TX-Box is the problem.	1. Check sequence parameters. Use only Siemens Sequences. 2. Check the EventLog of the system for warnings and errors of the cooling system, since these may be the root cause for the Tx-Box error message. 3. Check pressure at pressure reducer which is located at water inlet of TX-Box (approx. 3.2 bar), dis-connect (only some seconds) water input of TX-Box, displayed pressure (<0,6 bar). 4. Check cooling water temperature under SeSo / Magnet&Cooling / Cooling Status (18 - 22 degC) 5. Check RFCEL air cooling for blocking, leakage and for overtemperature under SeSo / Magnet&Cool-ing / Cooling Status. 6. Perform visual inspection of Tx coaxial cables and connectors (BC0, BC1, LC) for defects, loose or burned connectors. 7. Run QA / BC Tuning 8. Run TestTools / Normal / TX and Stability / TX. 9. If problem persists, replace TX-Box.
MRI_TXB [4] TxK2304 unit error: Tx-Box **gate volt-age max error** without rf opera-tion.	Mostly, TX-Box is the problem.	Check / replace TX-Box.

Error message/ [Error code]	Possible Reason	Action
MRI_TXB [6] TxK2304 unit error: Tx-Box cooling situation out of spec **(TXB_AIR_COOLANT_TEMPERATURE_ERROR)**.	Mostly, a cooling water problem: ■ cooling water over-temperature. Only in rare cases, TX-Box is the problem.	1. Check sequence parameters. Use only Siemens Sequences. 2. Check the EventLog of the system for warnings and errors of the cooling system, since these may be the root cause for the Tx-Box error message. 3. Check pressure at pressure reducer which is located at water inlet of TX-Box (approx. 3.2 bar), disconnect (only some seconds) water input of TX-Box, displayed pressure (<0,6 bar). 4. Check cooling water temperature under SeSo / Magnet&Cooling / Cooling Status (18 - 22 degC). 5. Check RFCEL air cooling for blocking, leakage and for overtemperature under SeSo / Magnet&Cooling / Cooling Status. 6. Run TestTools / Normal / TX and Stability / TX. 7. If problem persists, replace TX-Box.
MRI_TXB [31] TxK2304 unit error: An error in the high voltage power supply module occurred **(POSUMO_ERROR)**	Mostly TX-Box problem.	Check / replace TX-Box.

Error message/ [Error code]	Possible Reason	Action
MRI_TXB [32] TxK2304 unit error: RF System **Rectifier Error**	This error points to the THYRE or AC-supply problem.	For troubleshooting, see section (→ THYRE / Page 35).
MRI_TXB [31] TxK2304 unit error: An error in the high voltage power supply module occurred **(POSUMO_ERROR)** IN CONJUNCTION WITH **MRI_TXB [32]** TxK2304 unit error: RF System **Rectifier Error**	This error points to the THYRE or AC-supply problem. Only in rare cases, TX-Box is the problem.	For troubleshooting, see section (→ THYRE / Page 35).

2.3 THYRE

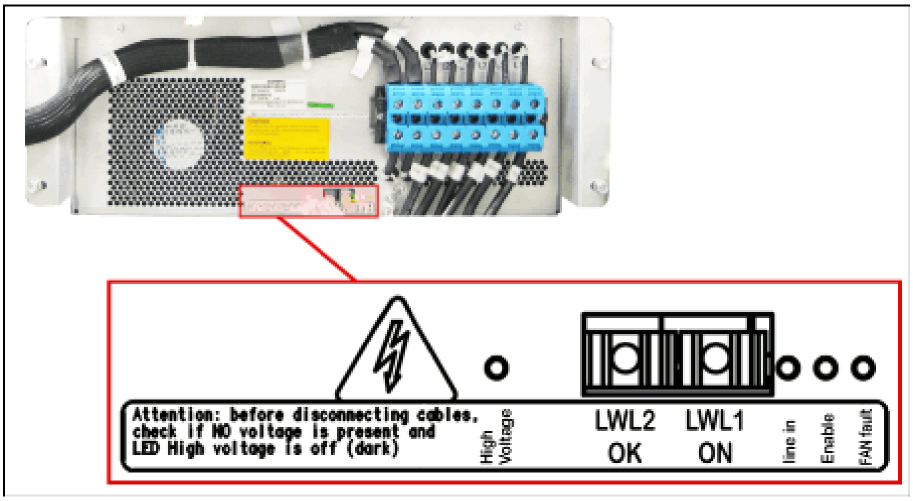
The description is valid for THYRE material numbers:

- 10432758 (THYRE1, 130 V for TXB_1)
- 10681869 (THYRE2, 110 V output for TXB_2)
- 10856740 (THYRE3, 110 V output for TXB_2CH 3T)

2.3.1 LEDs and FOCs

2.3.1.1 Overview

Fig. 16: Location of THYRE LEDs + FOCs



Tab. 10 LED description

LED	Description
High voltage	is connected to the 110 V output (THYRE2 and THYRE3), or 130 V (THYRE1). LED shows a variation of intensity, depending on output voltage. So, by discharge of the capacitor battery its brightness de-scends.
Line in	is connected to the internal 5V. If it doesn't light up, the reasons might be: ▪ Primary circuit breaker has tripped/off ▪ internal power supply unit defect
Enable	indicates 2.5 MHz signal, present at LWL1, i.e. starting command from TX-Box Controller (TBC).
FAN fault	indicates missing rotation feedback pulses from cooling fan, so it doesn't work. This error forces a shutdown.

Tab. 11 FOC description

FOC	Description
LWL1 ON	Switch on signal from the TX-Box controller (TBC). At beginning of boot procedure, LWL1 must be dark. During boot up procedure LWL1 will light up red for about 15 sec. to start THYRE. Pulling out the fiber-optic cable leads to a THYRE shutdown.
LWL2 OK	THYRE status signal to the TX-Box controller (TBC). Lights red if the starting command from TBC is set and the output voltage is in the proper range. Pulling out this fiber-optic cable leads to a THYRE shutdown, because the TBC doesn't receive the "ready" status and switches off the supply chain.

2.3.1.2 Normal Operation

Tab. 12 Normal Operation LED status

Mode	LED High voltage	LED Line in	LED Enable	LED FAN fault
Standby	off Capacitor voltage is down	ON internal power present	off TBC has not yet switched on	off FAN is checked after power up and has no error
Scanner Booting	LED lights up a few seconds after Enable is on	ON	Lights up at a certain moment in boot up sequence	off
Scanner Ready	ON Capacitors charged	ON	ON TBC has switched on	off

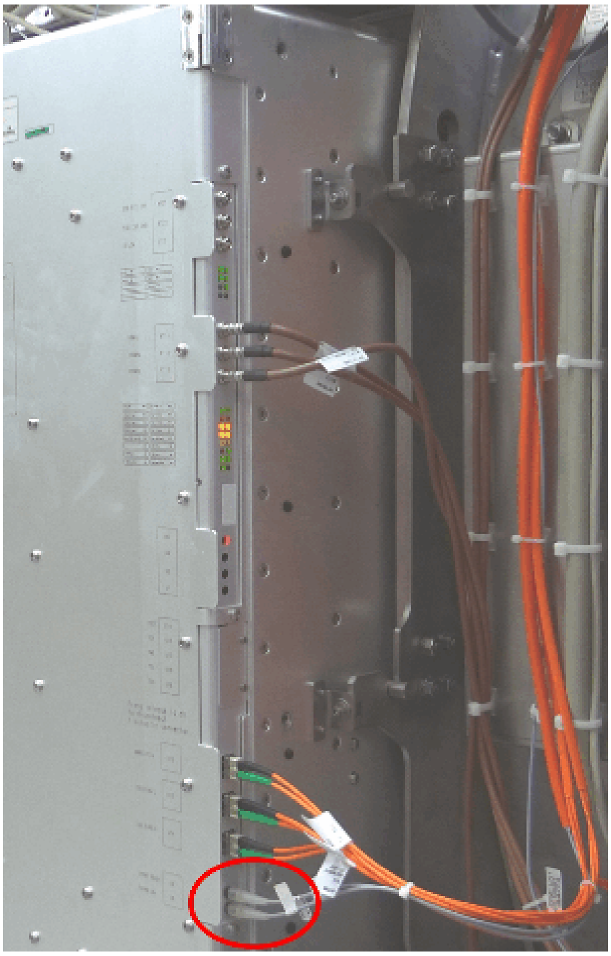
2.3.1.3 Error Conditions

Tab. 13 Error conditions LED status

<table>
<tr><th>Mode</th><th>LED
High voltage</th><th>LED
line in</th><th>LED
Enable</th><th>LED
FAN fault</th></tr>
<tr><td>Circuit breaker F51/F52 re-leased</td><td rowspan="3">off / ON
a certain time after shutdown this LED might light, even if the power supply unit is off (ca-pacitor dis-charge time).</td><td rowspan="3">off</td><td rowspan="3">off</td><td rowspan="3">off</td></tr>
<tr><td>Mains voltage error</td></tr>
<tr><td>Internal power supply fault</td></tr>
<tr><td>FAN fault</td><td>off / ON</td><td>ON</td><td>off</td><td>ON</td></tr>
<tr><td>THYRE error or Bias error (might be a result of TXB RAMBO error or POSUMO error)</td><td>off</td><td>ON</td><td>ON
at a certain mo-ment of boot up sequence, the TBC will switch on the THYRE and check for “ready” status af-ter 20sec.. If it is not ready, the enable signal will be canceled.</td><td>off</td></tr>
</table>

2.3.2 Check for correct start-up THYRE

Fig. 17: FOCs THYRE ON U6 / THYRE READY U9



Step 1:

- Disconnect fiber-optic cable connector U6 THYRE_ON from TBX, (see left figure).
- Reboot System
- At beginning of boot procedure, U6 may be either dark or light up fully bright.

  `Note:`

  `Initial U6 light status is controller random - just ignore! Also, ignore short reset, e.g.: full bright -> off -> full bright).`
- During boot up procedure, U6 light status must change to dimmed brightness for about 15 sec. (dimmed red = 2.5 MHz switch-on signal)
  - » TBX tries to switch ON THYRE, TBX function is OK
  - » If U6 status does not change to dimmed brightness, i.e. stays constantly dark, or fully bright, troubleshoot TBX

Step 2: (only if Step 1 successful)

- Reconnect U6 and disconnect FOC LWL2 OK at THYRE and reboot scanner.
- During boot up, first the Enable LED at the THYRE turns on. Then, with a delay of approx. 15 sec., LWL2_OK will light up red for a few seconds and switch off again. Reboot must fail.

» TBX switched ON THYRE successfully, and THYRE indicates READY

» If LWL2_OK stays dark and does not respond to the switch-on signal, troubleshoot THYRE, i.e. power supply unit.

2.3.3 Check of Supply and Output Voltages THYRE

WARNING

Dangerously high voltage can be present even when the MR system is off.

Risk of electrical shock!

» **Before start working at the THYRE or TX-Box, switch off the power for the TX-system (circuit breaker F51/ F52 in EPC).**

» **Wait at least 10 min. to discharge capacitors and check for no voltage at the THYRE output.**

» **Do not remove any cables before the high voltage LEDs at THYRE and the TX-Box are off.**

WARNING

THYRE voltages are floating ! Do not ground floating voltages with measurement equipment !

Risk of electrical shock or severe injury!

» **Only use potential free and fully insulated measurement equipment like a DVM.**

» **DO NOT USE any measurement equipment which is connected to a power supply unit or ground (e.g. standard oscilloscope).**

NOTICE

THYRE thyristor switching can cause AC supply voltage distortions.

Wrong DVM voltage readings in case the THYRE is enabled during DVM measurements !

» **Always disable THYRE before measuring AC supply (input) voltages.**

» **Pull out FOC LWL1_ON to disable THYRE prior to DVM measurements.**

Two types of EPC transformer with different winding configuration are used (mdexx Dy5d4 or Noratel Dy5d6).

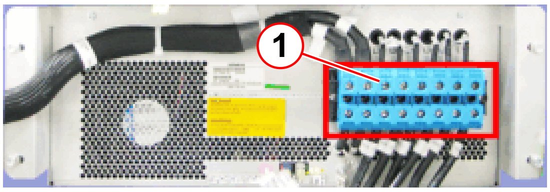
The AC supply voltage for THYRE is different for both transformer versions !

Since the transformer label is not easily accessible, check the specific AC supply voltages for THYRE according to the table below.

- The THYRE is supplied with 6 Phases from the EPC mains voltage transformer (circuit breakers F51/F52).
- Pull out FOC LWL1_ON at the THYRE to disable the unit.

- Measure the THYRE AC supply (input) voltages with a potential free and fully insulated measurement instrument (e.g. DVM).

Tab. 14 AC supply voltages THYRE

<table>
<tr><th>Test Points</th><th>Voltage for EPC main transformer type 1
(mdexx Dy5d4)</th><th>Voltage for EPC main transformer type 2
(Noratel Dy5d6)</th><th>Action</th></tr>
<tr><td></td><td colspan="2">Fig. 18: THYRE


(1) Voltage Test Points</td><td></td></tr>
<tr><td>L1 - L2</td><td rowspan="6">113 + 2 / - 1 V

(112 - 115 V is acceptable)</td><td rowspan="6">115 ± 1 V

(112 - 115 V is acceptable)</td><td rowspan="15">1. **In case voltages are far too low or missing, check:**
- Mains voltage
- System main contactor
- EPC main transformer
- Circuit breaker F51/F52
2. **In case voltages are too low but within a tolerance band of +/- 15 V, check:**
- Mains voltage for Total Harmonic Distortion (THD) with PQube measurement device
- THD must be < 5%
- Contact HSC for further instructions</td></tr>
<tr><td>L2 - L3</td></tr>
<tr><td>L1 - L3</td></tr>
<tr><td>L11 - L21</td></tr>
<tr><td>L21 - L31</td></tr>
<tr><td>L11 - L31</td></tr>
<tr><td>L1 - L11 (4U2-3U2)</td><td rowspan="3">33 ± 2 V</td><td rowspan="3">33 ± 2 V</td></tr>
<tr><td>L2 - L21 (4V2-3V2)</td></tr>
<tr><td>L3 - L31 (4W2-3W2)</td></tr>
<tr><td>L1 - L21 (4U2-3V2)</td><td rowspan="3">90 ± 3 V</td><td rowspan="3">123 ± 3 V</td></tr>
<tr><td>L2 - L31 (4V2-3W2)</td></tr>
<tr><td>L3 - L11 (4W2-3U2)</td></tr>
<tr><td>L1 - L31 (4U2-3W2)</td><td rowspan="3">123 ± 3 V</td><td rowspan="3">90 ± 3 V</td></tr>
<tr><td>L2 - L11 (4V2-3U2)</td></tr>
<tr><td>L3 - L21 (4W2-3V2)</td></tr>
</table>

- After measurements, reconnect FOC LWL1_ON to THYRE and reboot scanner.
- Check if the THYRE is up and properly operating. LEDs “Line In”, “Enable” and “High Voltage” must be on.
- Measure the THYRE DC output voltage with a potential free and fully insulated measurement instrument (e.g. DVM).

The THYRE DC output voltage must be measured with an enabled THYRE.
LEDs “Line In”, “Enable” and “High Voltage” must be on.

Tab. 15 DC output voltage THYRE

Test Points	Voltage	Action
THYRE.Xx.+ - THYRE.Xx.-	108 V ... 112 V DC (for TXB_2) 128 V ... 132 V DC (for TXB_1) **Note:** The voltages are measured with the system switched on and no sequence running.	■ If voltage is outside specification, perform complete check of (→ THYRE / Page 35) before replacing the unit.

In case of THYRE problems, also check cabling (bad contacts, loose screws?).

- All cable connections at THYRE AC-supply side:
  - EPC Main Transformer
  - Circuit Breakers F51, F52
  - THYRE terminals, see (→ Pos.1/Fig. 18 Page 40).
- All cable connections at THYRE DC-output:
  - THYRE terminals (THYRE.Xx.+ / THYRE.Xx.-)
  - RF Filter Plate (X40 (equipment room) - Filter Z20, Z21 - X22A (magnet room))
  - TX-Box connector, see (→ Fig. 7 Page 18).

2.3.4 Special Problems

The following section contains additional troubleshooting procedures for specific THYRE problems.

Problem	Possible Reason	Action
THYRE Circuit Breaker F51/F52 has tripped-off (specially, during start-up of TX-Box)	THYRE AC supply voltage phase(s) missing.	Check THYRE supply voltages, see (➔ Check of Supply and Output Voltages THYRE / Page 39).
	The THYRE AC supply voltage is distorted by harmonics.	Check for Total Harmonic Distortion (THD) < 5%, see (➔ Check of Supply and Output Voltages THYRE / Page 39).
	Short circuit at THYRE DC output.	Check for short circuit at THYRE DC output. ▪ Select **System/ Control/ Meas Recon/ Standby** ▪ At the EPC, switch off circuit breakers F51/F52/ THYRE and F10/ RFCEL/TX-Box. ▪ Wait at least 10 min. and **check for NO VOLTAGE** at the THYRE output. ▪ Check that the high voltage LEDs at THYRE and the TX-Box are off. ▪ Unplug the Power Line Filter (PoliFi) plug from the left side of the TX-Box ▪ Disconnect both DC output cables at terminals THYRE.Xx.+ and THYRE.Xx.-, see figure (➔ Fig. 18 Page 40). ▪ Check with a DVM in resistance mode for short circuit of cable leads to TX-Box and filters in the RF-panel. Also measure against ground. ▪ Reconnect the Power Line Filter (PoliFi) plug to the TX-Box and check with DVM for short circuit of the TX-Box.

2.4 Switch_MNO

The function of the Switch_MNO can only be tested by setting the system into definite operation modes and checking the correct function and the LEDs of the switch.

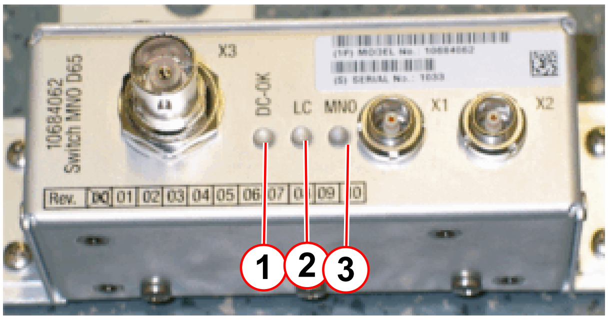
The normal states as well as possible error conditions are shown in the following table.

> Before troubleshooting MNO problems, check for correct cabling.
> Confused cables (e.g. swapped LO cables at IF_RCCS) are not detected by the Software.

2.4.1 Function Test

- Check the state of the LEDs on the Switch_MNO (located at TX-Box left-side):

Fig. 19: MNO-Switch LEDs



(1) LED DC- OK (green)
(2) LED LC (yellow)
(3) LED MNO (yellow)

Regular conditions:			
Mode	**LED DC-OK (green)**	**LED LC (yellow)**	**LED MNO (yellow)**
BC Imaging	**On** DC supply from RFCEL_SAT/X19 via cable W15903 to MNO-Switch/X3 ok	**Off** 3 VDC level on cable W15900 from RCCS/ BC1 to MNO-Switch/X2 ok	**Off** LO signal on cable W15900 from RCCS/ BC1 to MNO-Switch/X2 ok
LC Imaging	**On**	**On** No 3 VDC level on cable W15900 from RCCS/ BC1 to MNO-Switch/X2	**Off** LO signal on cable W15900 from RCCS/ BC1 to MNO-Switch/X2 ok

Regular conditions:			
Mode	**LED DC-OK (green)**	**LED LC (yellow)**	**LED MNO (yellow)**
MNO	**On**	**On** No 3 VDC level on cable W15900 from RCCS/ BC1 to MNO-Switch/X2	**On** No LO signal on cable W15900 from RCCS/ BC1 to MNO-Switch/X2
LED test MNO-Switch X1 and X2 disconnected	**On**	**On**	**On**

Error conditions:			
Mode	**LED DC-OK (green)**	**LED LC (yellow)**	**LED MNO (yellow)**
All modes	**Off** Check DC supply from RFCEL_SAT/X19 via cable W15903 to MNO-Switch/X3	-	-
BC Imaging	**On**	**On** Missing 3 VDC level on cable W15900 from RCCS/ BC1 to MNO-Switch/X2 Check cable W15900 from RCCS/ BC1 to MNO-Switch/X2, if ok -> Error RCCS	**Off**
	On	**Off**	**On** LO signal from RCCS BC1 to MNO-Switch/X2 is missing Check cable W15900 from RCCS/ BC1 to MNO-Switch/X2 Connect cable between synthesizer LO_H and MNO-switch/X2: LED must go Off.

Error conditions:			
Mode	**LED DC-OK (green)**	**LED LC (yellow)**	**LED MNO (yellow)**
LC Imaging	**On**	**Off** 3 VDC level on cable W15900 from RCCS/ BC1 to MNO-Switch/X2 -> Error RCCS	**Off**
	On	**On**	**On** LO signal from RCCS BC1 to MNO-Switch/X2 is missing Check cable W15900 from RCCS/ BC1 to MNO-Switch/X2 Connect cable between synthesizer LO_H and MNO-switch/X2: LED must go Off.
MNO	**On**	**On**	**Off** LO signal on cable W15900 from RCCS/ BC1 to MNO-Switch/X2
	On	**Off** 3 VDC level on cable W15900 from RCCS/ BC1 to MNO-Switch/X2 -> Error RCCS	**On**
LED Test MNO-Switch X1 and X2 disconnected	**On**	**If Off** -> defective LED	**If Off** -> defective LED

Version 05:

- Products MAGNETOM **Prisma** and **Prisma**^{fit} added.
- Section (TX-Box), added procedure (→ Measurement of load condition DC Voltages at RF panel feed-through filters / Page 21) (CHARM_MR_00432972).
- Section (→ THYRE / Page 35), DC supply voltages for different versions corrected (110/ 130 VDC). (CHARM_MR_00445313)
- Section (THYRE), corrected procedure (→ Check of Supply and Output Voltages THYRE / Page 39). AC supply (input) voltages are to be measured with disabled THYRE.
- Section (THYRE), corrected procedure (→ Check for correct start-up THYRE / Page 38). Correction of status FOC U6 THYRE_ON during scanner boot.

Version 06:

- Section (TX-Box), corrected table (→ Tab. 9 Page 28). Changed system serial number bands for LC-TX cable extension and regulator error removed (CHARM MR_00461639).
- Section (TX-Box), complete rework of LED description, TX-Box1 added.
- Section (TX-Box), error correction in graphics (→ Fig. 9 Page 18) (Document Feedback - swapped connectors X26 and X28).
- All sections, minor editorial changes.

Version 07:

- Section (TX-Box), new chapter (→ Resolving Feedback Loop Errors / Page 24) added.
- Section (TX-Box), chapter (Special Problems), added common TX-Box errors to troubleshooting table (→ Tab. 9 Page 28).
- Section (→ THYRE / Page 35), added hint to check THYRE cabling and contacts in case of voltage problems.

Version 08

- Section (General information), chapter (Requirement (→ Additionally Required Documents / Page 8)) error messages can be found in the SKB added.

Version 09

- Section (TX-Box), chapter (Special Problems), added additional action to error: MRI_TBC [220], MRI_TXB [2] and MRI_TXB [6]

  Charm MR_00484717
- Section (Thyre) minor editorial changes.

There are no Hazard IDs in this document.

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